

HMITiles

An Open-Source HMI Tile Library for Industrial Dashboards

Overview

HMITiles is an open-source HMI (Human Machine Interface) tile library written in **B4X**, following widely accepted industrial HMI principles.

It provides reusable, professional-grade HMI tiles for: Industrial dashboards, SCADA (Supervisory Control and Data Acquisition) front-ends, Machine and process HMIs.

The focus of this library is **clarity, consistency, and disciplined HMI design** - not visual effects or UI decoration.

This is a **non-commercial project** shared for inspiration.

B4J Example – Overview HMITiles

HMITiles Example Overview v20251231 (B4J)

Label Normal Label Warning Label Alarm 11:54:10 Button Toggle Button Alarm Button OnOff

Sensor Warning Motion SeekBar Indicator Gauge

85°C Value Detected 85 85 85

RGB Hor RGB Vert Level Sensor Trend Image

20 2 Select Select

Setpoint Setpoint Item 1 ON Label Half

25 5 Item 2

Δ -5 Item 3

Event Viewer

11:54:04 - [TileSeekBar_ValueChanged] value=85

11:53:52 - [TileSelect_ValueChanged] value=ON

11:53:52 - [TileSelect_ValueChanged] value=ON

11:53:52 - [TileSPPV1_SetPointChanged] value=25.0

List

item 0

0

item 1

1

item 2

2

item 10

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HMITiles Used
 HMITileLabel (3)
 HMITileDigitalClock
 HMITileClock
 HMITileButton (3)

HMITileSensor (3)
 HMITileSeekBar (2)
 HMITileGauge

HMITileRGB (2)
 HMITileLevel
 HMITileReadOut
 HMITileTrend
 HMITileImage

HMITileSetPoint
 HMITileSelectList
 HMITileSelect
 HMITileShape (8)
 HMITileImageIcon

HMITileEventViewer
 HMITileList

B4J Example - Water Tank Simulator

A simple interactive water tank PID simulator demonstrating selective HMITiles.

HMITiles Example Water Tank Simulator v20251231

Input

FIC-101
38.2
- 45 +
Δ -6.8

Tank Level

FR-101


LIC-101
67
50
Δ 17

LR-101


Output

FI-102
39.3

FR-102


Simulator
ON

VPI-102
42
Valve %


VPR-102


Event Viewer

 11:56:14 - [SimulatorStep] Input=38.2, Level=67, Output=39.3

 11:56:14 - [TileSPPVTankLevel_ValueChanged] value=67.0

 11:56:12 - [SimulatorStep] Input=35.2, Level=69, Output=35.7

 11:56:12 - [TileSPPVTankLevel_ValueChanged] value=69.0

 11:56:10 - [SimulatorStep] Input=30.8, Level=70, Output=24.1

 11:56:10 - [TileSPPVTankLevel_ValueChanged] value=70.0



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HMITiles Used

- HMITileLabel
- HMITileSetpoint
- HMITileTrend
- HMITileSetpoint
- HMITileReadOut
- HMITileTrend
- HMITileButton
- HMITileSeekBar
- HMITileTrend
- HMITileEventViewer

B4J Example – ArduinoLED

Control the state of an LED connected to an Arduino UNO via the serial line.

HMITiles Used

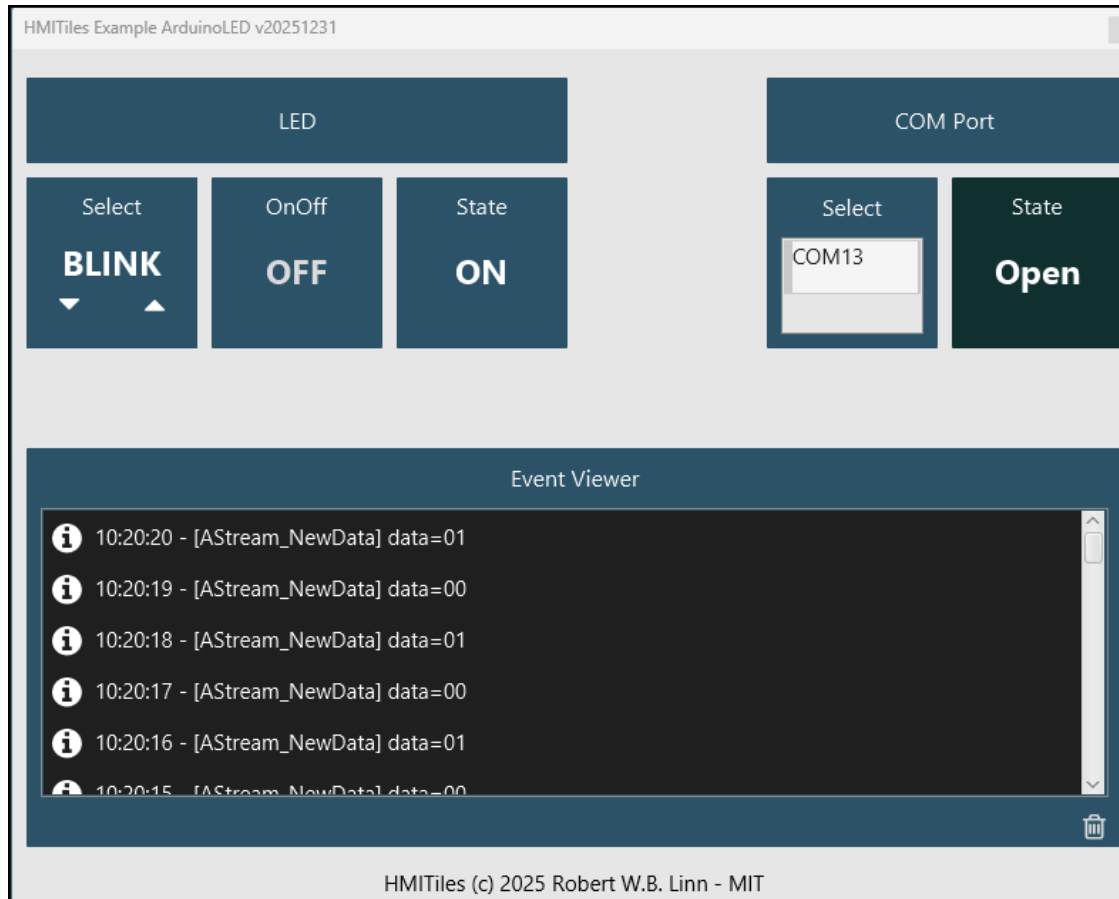
HMITileLabel

HMITileSelect

HMITileButton

HMITileReadOut

HMITileEventViewer



HMITiles Used

HMITileLabel

HMITileSelectList

HMITileButton

HMITileEventViewer

B4A Example – Overview



HMITiles Used

HMITileLabel (3)
HMITileDigitalClock
HMITileClock

HMITileButton (3)
HMITileTrend
HMITileGauge

HMITileSensor
HMITileReadOut
HMITileLevel

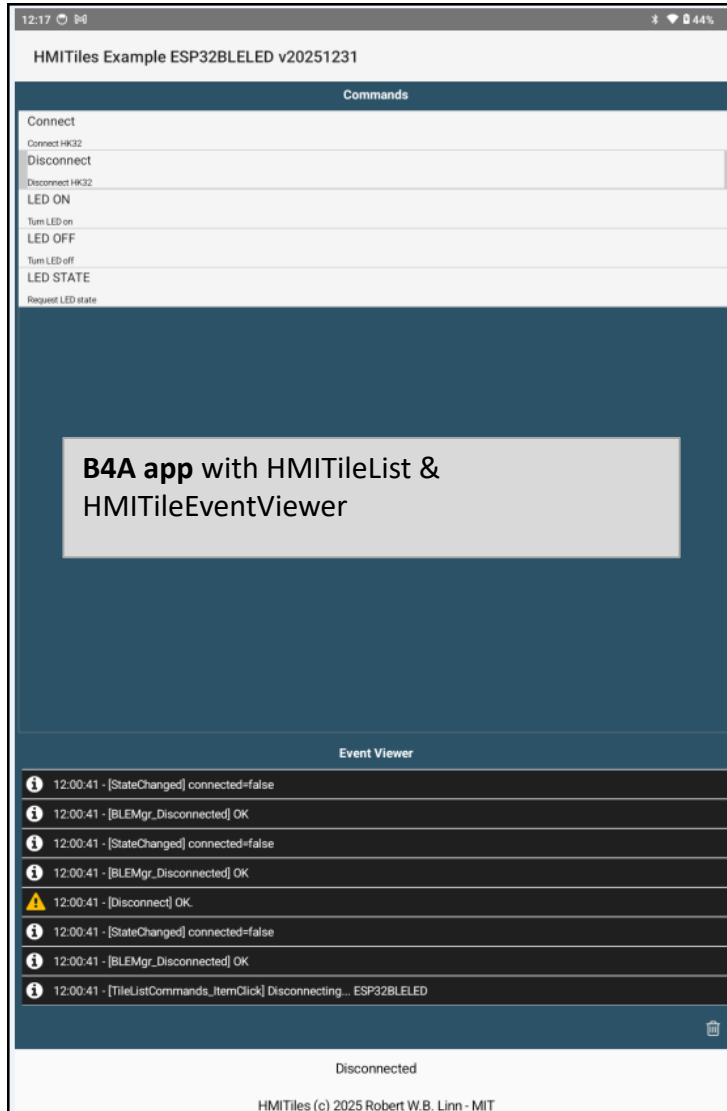
HMITileSeekBar (2)

HMITileSetPoint
HMITileSelectList
HMITileSelect

HMITileEventViewer

B4A Example – ESP32BLELED

Control the state of an LED connected to an ESP32 via the BLE (UART).

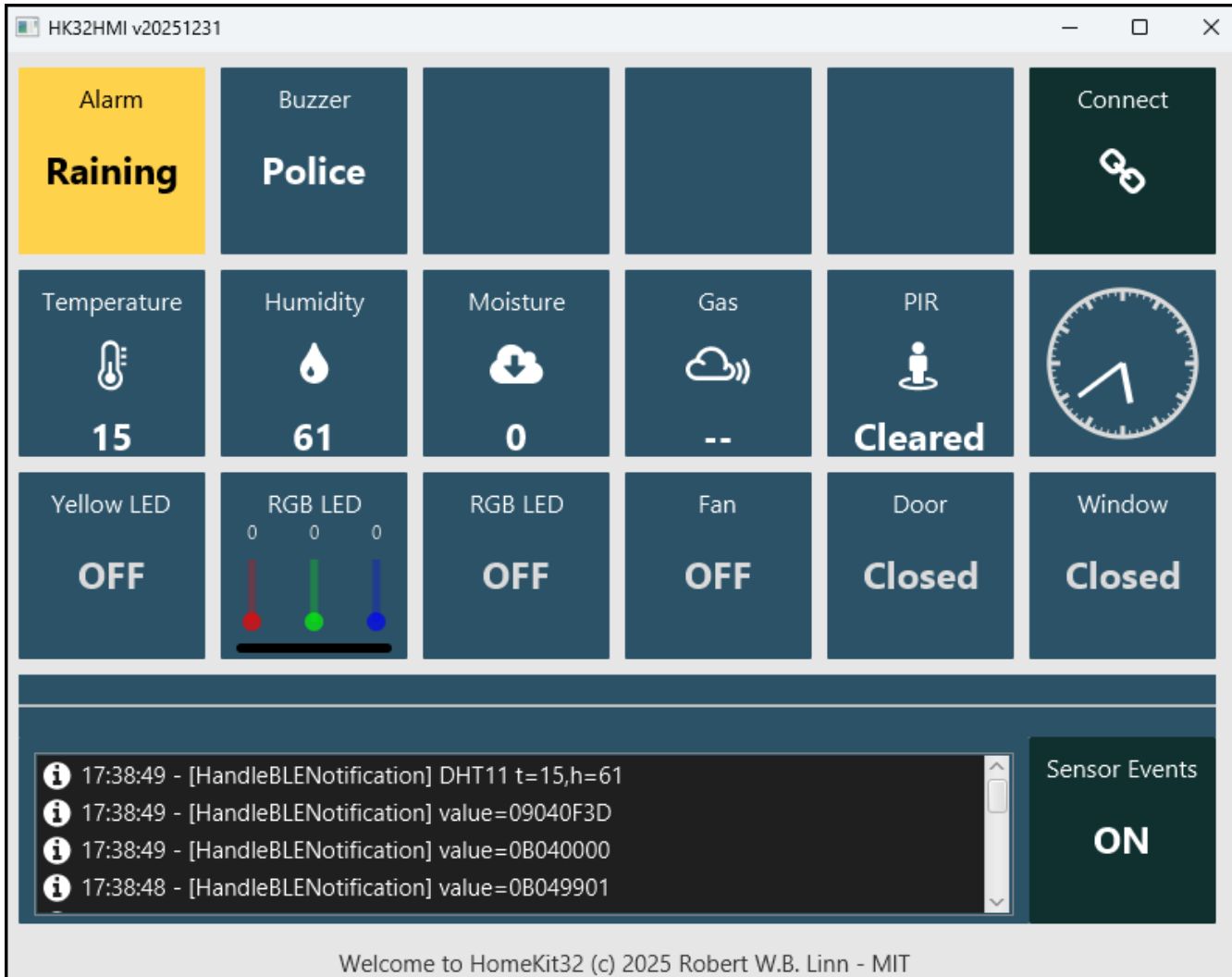


```
[B4RBLEServer::onConnect] Client connected
[CommBLE.BLEServer_NewData] buffer HEX=010101
[CommBLE.BLEDispatch] deviceid=1
[DevLED.ProcessBLE] payload=010101
[CommBLE.BLEServer_Write] data=010101
[B4RBLEServer::Write] Notified bytes: 3
[DevLED.WriteToBLE] payload=010101
[B4RBLEServer::onWrite] Received bytes: 3
[CommBLE.BLEServer_NewData] buffer HEX=010100
[CommBLE.BLEDispatch] deviceid=1
[DevLED.ProcessBLE] payload=010100
[CommBLE.BLEServer_Write] data=010100
[B4RBLEServer::Write] Notified bytes: 3
[DevLED.WriteToBLE] payload=010100
[B4RBLEServer::onWrite] Received bytes: 3
[CommBLE.BLEServer_NewData] buffer HEX=0102
[CommBLE.BLEDispatch] deviceid=1
[DevLED.ProcessBLE] payload=0102
[CommBLE.BLEServer_Write] data=010200
[B4RBLEServer::Write] Notified bytes: 3
[DevLED.WriteToBLE] payload=010200
[B4RBLEServer::onWrite] Received bytes: 2
[B4RBLEServer::SetStartAdvertising] Started advertising: ESP32BLELED
[B4RBLEServer::onDisconnect] Client disconnected, restarting advertising
```

B4R program IDE log showing the commands:
connect > command LED ON > command LED OFF > disconnect

B4J Example – HomeKit32

HomeKit32 is a modular smart-home automation framework for the Keyestudio Smart Home Kit (KS5009), powered by **ESP32, BLE, MQTT and B4X**.



HMITiles Used

HMITileLabel (2)
HMITileButton

HMITileSensor (4)
HMITileDigitalClock
HMITileClock

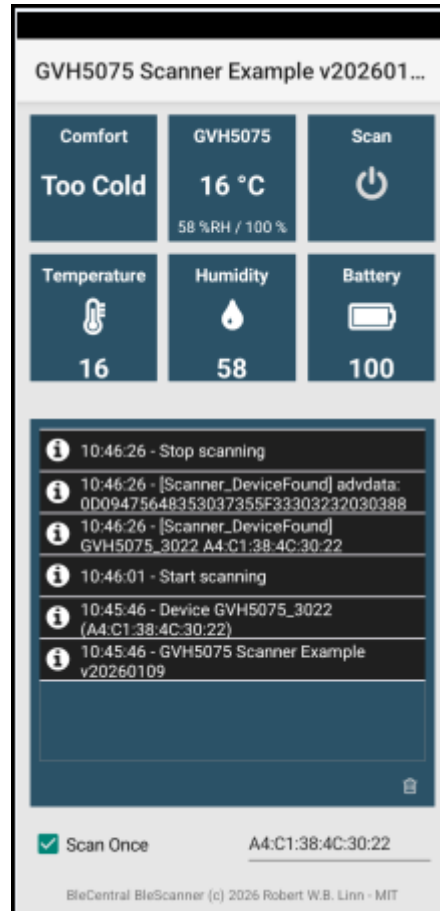
HMITileButton (5)
HMITileRGB

HMITileEventViewer
HMITileButton



B4A Examples – BLE GVH5075 & RuuviTag

B4A additional library **BleCentral** examples scanning devices Govee GVH5075 & RuuviTag.



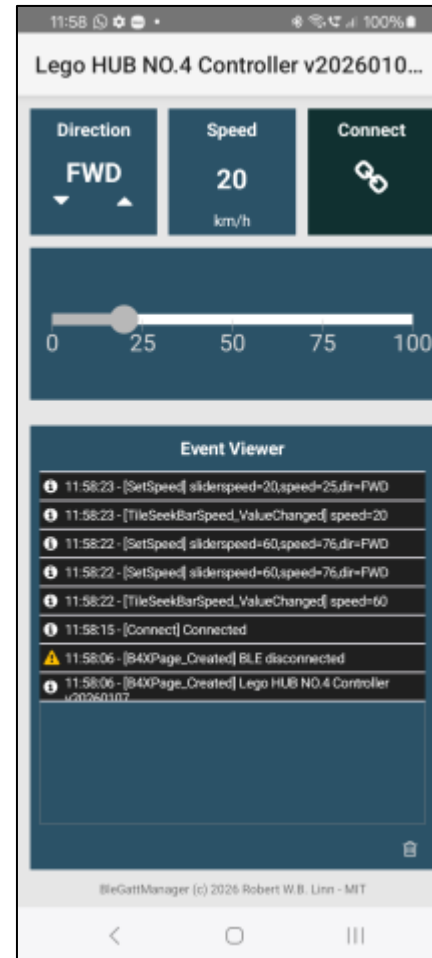
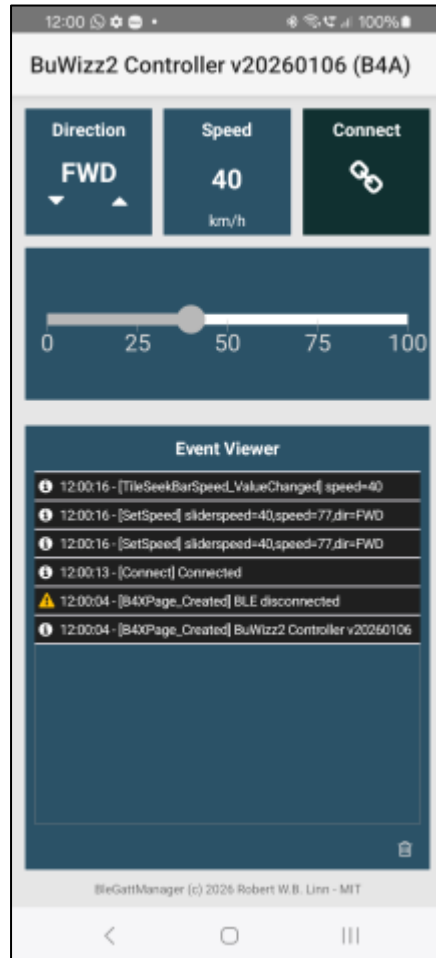
HMITiles Used

HMITileButton
HMITileSensor

HMITileEventViewer

B4A Examples – BLE BuWizz2 & Lego HUB No.4

B4A additional library **BleCentral** examples controlling bricks BuWizz2 & Lego HUB No.4.



HMITiles Used

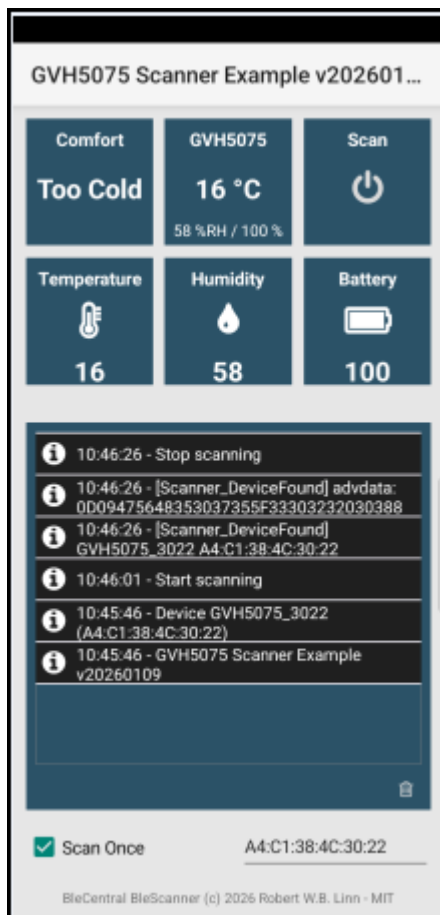
HMITileSelect
HMITileReadOut
HMITileButton

HMITileSeekBar

HMITileEventViewer

B4A Examples – BleCentral.b4xlib

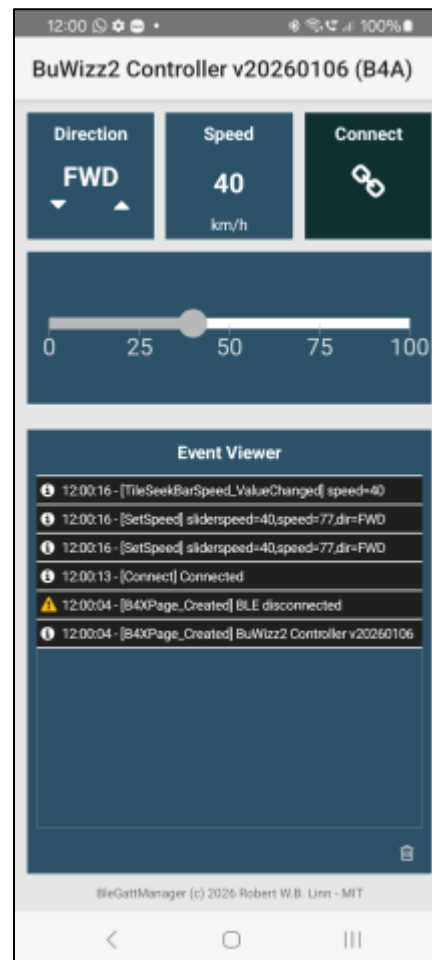
HMITiles B4A examples using the B4A library BleCentral.b4xlib



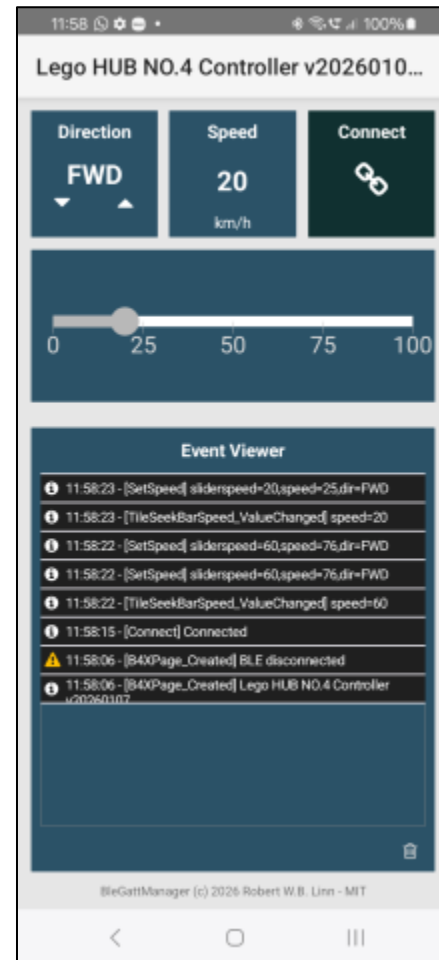
Govee Bluetooth hygrometer and thermometer.



Open-source Bluetooth sensor temperature, humidity, air pressure, and motion.



Compact, rechargeable battery and Bluetooth remote control brick for Lego models.



Lego Powered Up Bluetooth Hub.

Screenshots